

# Edeon — Registration Readiness Dossier

**Compound:** Imidacloprid (CAS: 138261-41-3)

**SMILES:** C1=C(C1)C=NC(=C1)CN2C(=N[N+](=O)[O-])NCC2

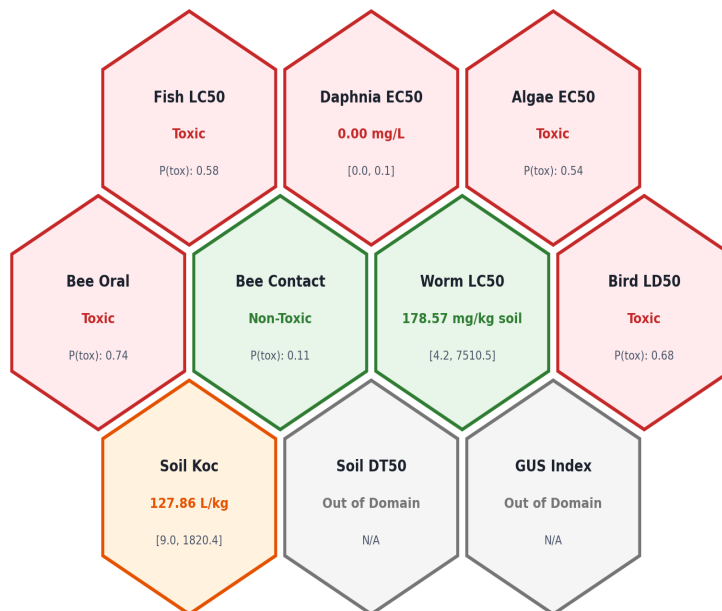
## REGISTRATION READINESS VERDICT: NO\_GO

**Driver:** Showstopper triggered: Groundwater concern (EU DWD 0.1 µg/L trigger), Aquatic hazard (CLP H400/H410/H411/H412), Soil persistence (EU 1107/2009) | **Confidence:** moderate

**Rationale:** GUS = 4.21 > 2.8 (leacher), GUS classification: leacher; Most sensitive endpoint: Daphnia EC50 = 0.0000 mg/L, Not rapidly degradable (default assumption); Predicted soil DT50 = 168 days, DT50 > 120 d — exceeds persistence cut-off criterion

## Tier-1 Predictive Honeycomb

### Imidacloprid — QSAR Safety Profile



## Regulatory Cut-Offs & Criteria Evaluation

| Criterion / Endpoint                               | Status                         | Evidence & Details                                                                               |
|----------------------------------------------------|--------------------------------|--------------------------------------------------------------------------------------------------|
| Genotoxicity / Mutagenicity alerts (Benigni-Bossa) | <b>WATCH</b>                   | Nitroamine (N-nitro) (medium): Metabolic reduction to reactive intermediates                     |
| Endocrine disruption screening (structural)        | <b>PASS</b>                    | No endocrine disruption structural alerts detected                                               |
| Skin sensitization alerts (protein reactivity)     | <b>PASS</b>                    | No skin sensitization alerts detected                                                            |
| PBT/vPvB (REACH Annex XIII)                        | <b>WATCH</b>                   | Soil DT50 = 168 d > 120 d (P); Daphnia EC50 = 0.0000 mg/L < 0.1 (T)                              |
| Groundwater concern (EU DWD 0.1 µg/L trigger)      | <b>LIKELY_SHO<br/>WSTOPPER</b> | GUS = 4.21 > 2.8 (leacher); GUS classification: leacher                                          |
| Aquatic hazard (CLP H400/H410/H411/H412)           | <b>LIKELY_SHO<br/>WSTOPPER</b> | Most sensitive endpoint: Daphnia EC50 = 0.0000 mg/L; Not rapidly degradable (default assumption) |
| Acute mammalian toxicity (CLP oral)                | <b>PASS</b>                    | Predicted rat oral LD50 = 10399 mg/kg                                                            |
| Soil persistence (EU 1107/2009)                    | <b>LIKELY_SHO<br/>WSTOPPER</b> | Predicted soil DT50 = 168 days; DT50 > 120 d — exceeds persistence cut-off criterion             |

*IN-SILICO SCREENING ONLY — These results are computational triage signals based on predicted endpoints and structural pattern matching. They are NOT regulatory determinations and cannot replace experimental studies or expert regulatory assessment.*